QUANT

Mensuration & Geometry

(Circles, Quadrilaterals & Trigonometry)

Assignment 02

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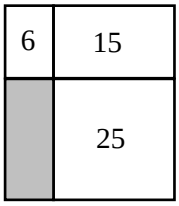
Geometry Online Assignment Cluster II -02

1. A rectangle is divided into 4 smaller rectangles. The areas of three of these rectangles are 6 cm^2, 15 cm^2 and 25 cm^2 as shown. What is the ratio of the sum of the perimeters of the four smaller rectangles to the perimeter of the original rectangle?

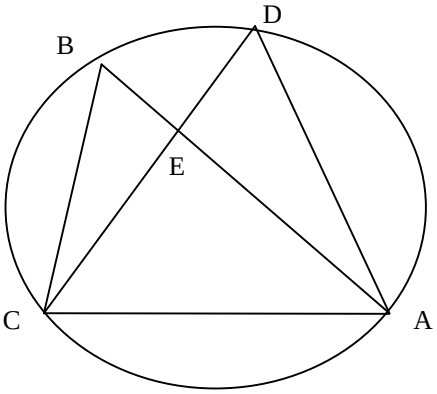


1. 2 : 1 2. 1 : 1 3. 3 : 2 4. Insufficient data

2 The radius of a circle is decreased by 20 %. What will be the percentage decrease in its surface area?

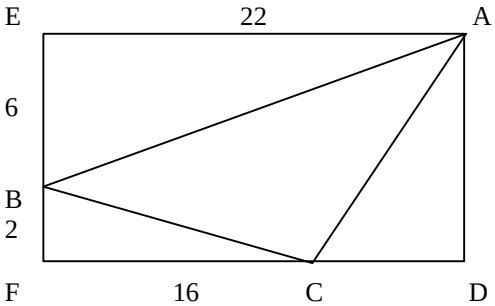
1. 64% 2. 36% 3. 20% 4. 10%

3. In the figure below, points A, B, C and D lie on the circle. AD = 24 and BC = 12. What is the ratio of the area of triangle CBE to that of triangle ADE?



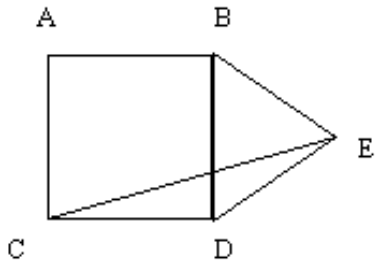
1. 1 : 4 2. 1 : 2 3. 1 : 3 4. 2 : 3

4. In the given figure EADF is a rectangle and ABC is a triangle whose vertices lie on the sides of EADF. AE = 22, BE = 6, CF = 16 & BF = 2. Find the length of the line joining the mid-points of the sides AB and BC.



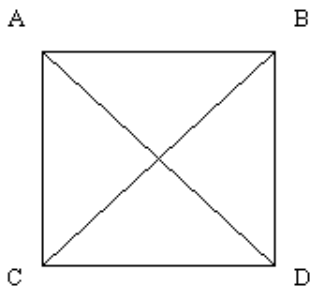
1. $4\sqrt{2}$ 2. 5 3. 3.5 4. 4.5

5. If ABDC is a square and BDE is an equilateral triangle, what is the measure of angle DEC?



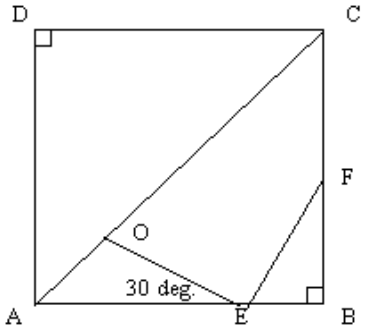
1. 15° 2. 30° 3. 20° 4. 25^0

6. ABCD is a square of area 4 which is divided into four non-overlapping triangles as shown in figure. Then sum of the perimeters of the triangles is



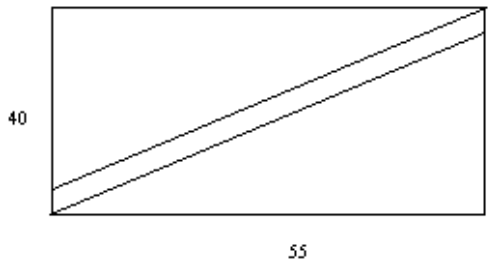
1. $8(2 + \sqrt{2})$ 2. $8(1 + \sqrt{2})$ 3. $4(1 + \sqrt{2})$ 4. $4(2 + \sqrt{2})$

7. AE = CF, EB = FB, what is the value of angle OEF ?



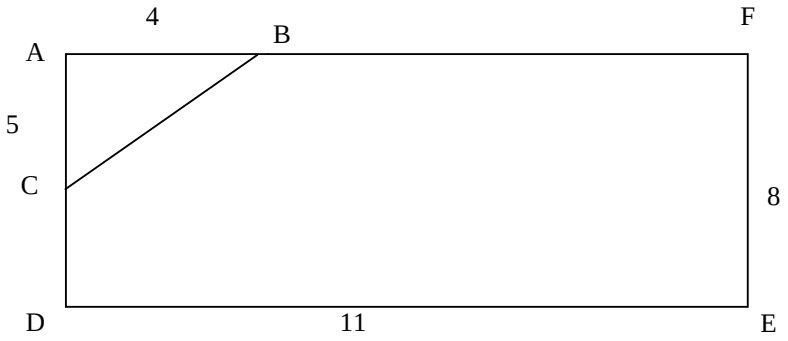
1. 90^0 2. 135^0 3. 105^0 4. 100^0

8. The figure shows a diagonal path across a rectangular garden measuring 55 m by 40 m. If the path is exactly 1 m wide, what is its area?



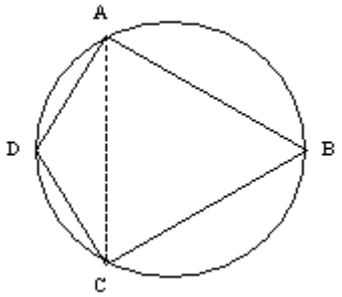
1. 100 sq m 2. 55 sq m 3. $112\frac{2}{3}$ sq m 4. $88\frac{1}{8}$ sq m

9. Corner ABC is cut from rect AFED. Find the remaining area?



1. 78 2. 88 3. 68 4. 72

10. ABCD is a cyclic quadrilateral such that ADB = 30 degrees DCA = 80 degrees. Then DAB =



1. 135 degrees 2. 100 degrees 3. 120 degrees 4. 70 degrees

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Answer key & Explanations

Q. No.	Ans.	Exp.
1.	1	We can observe that the perimeters of the four rectangles will add up to twice the perimeter of the original rectangle. The ratio is 2 : 1.
2.	2	$0.8 \times 0.8 = 0.64$. So 36 % decrease.
3.	1	We know that angles inscribed in the same arc are congruent. Using this we can show that triangle CBE is similar to triangle ADE. The ratio of the areas of these triangles is the same as the ratio of the squares of the corresponding sides. The ratio of sides happens to be 12 : 24 i.e. 1 : 2. The ratio of areas will become $1^2 : 2^2$ $\Rightarrow$ 1 : 4. Thus 1st option is the answer.
4.	2	From the figure it can be seen that length of CD = 22 – 16 = 6. Using Pythagoras theorem in triangle ADC, AC^2 $= AD^2 + CD^2 = 8^2 + 6^2 = 100$. So length of AC = 10. We know that the line joining the mid-points of two sides of a triangle is parallel to the third side and half the length of the third side. Therefore the length of the line joining the mid-points of AB and BC = $\frac{1}{2} \times 10 = 5$. Thus 2 is the answer.
5.	1	Since BDE is an equilateral triangle on one side of the square each side of triangle BDE will be equal to the side of the square. Thus DC = DE i.e. triangle DCE is an isosceles triangle in which $\angle$ DCE = $\angle$ CED. But $\angle$ EDC = $\angle$BDC + $\angle$ EDB = 90 + 60 = 150°. Thus $\angle$ DCE + $\angle$CED = 30° and both are equal also, hence both of these $\angle$s are equal to 15 degrees.
6.	2	Side of square = 2. Diagonal = $2\sqrt{2}$. Sum of perimeters of triangles = perimeter of square + 2(sum of diagonals) $= 4(2) + 2(4\sqrt{2}) = 8(1 + \sqrt{2})$. NOTE: Since 2 adjacent triangles share one side, the diagonals are added twice.
7.	3	Since EB = FB, angle BEF = angle BFE = 45°. Hence in segment AEB, 30 + angle OEF + 45 = 180. So angle OEF = 180 – 75 = 105°. Thus 3rd option is the answer.
8.	2	Area of the rectangle is 55 × 40 = 2,200 sq m. Area of the triangle = 55 meters base, 39 (40 – 1) meters height. Area of both triangles is 2,145 sq m. The area of the strip is = (area of the rectangle) – (area of two triangles). = 2,200 – 2,145 = 55 sq mts.
9.	1	Area = 88 - 10 = 78.
10.	4	From the figure BCA = 30. So DCB = DCA + BCA = 80 + 30 = 110. DAB = 180 – 110 = 70. Hence 5.